## Student & Internship Details

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|---|---|
| **Student Name** | Vinayak Dubey |
| **Programme & Batch** | PGDM \| Batch 2025-27 |
| **Organization Name** | Negbuy Private Limited |
| **Industry / Sector** | Procurement & Supply Chain |
| **Internship Duration** | 12 weeks |
| **Faculty Supervisor** | Dr. Gunjan Sood |
| **Organizational Supervisor** | Mr. Sayeem Salmani |
| **Date of Submission** | 11th June, 2026 |

## Section 1 — Problem Statement

### 1.1 What is the problem? Who does it affect, and how?

During the internship at Negbuy, it was observed that the procurement team regularly receives RFQs in different unstructured formats such as screenshots, product images, PDFs, Excel sheets, WhatsApp messages, and technical descriptions. Before supplier sourcing can begin, procurement personnel manually review these RFQs to identify part numbers, manufacturers, specifications, and product details.

A major issue within the current workflow is that procurement information and supplier references are not maintained through a centralized system. As a result, procurement personnel often repeat the same supplier search and quotation activities even for similar RFQs. This increases manual effort, slows down procurement execution, and makes supplier tracking difficult.

The problem mainly affects procurement interns, sourcing executives, and supply chain personnel who are directly involved in supplier identification, quotation handling, and RFQ management activities.

### 1.2 How is this problem currently being solved (without AI)?

At present, the entire RFQ handling and sourcing process is managed manually. The procurement team receives RFQs through email, WhatsApp, spreadsheets, screenshots, and product photographs. Team members manually extract part numbers and manufacturer details before searching for suppliers on Alibaba, Google, and distributor websites.

For standard industrial products, supplier sourcing is generally done through Alibaba and similar platforms. For specialized industrial and turbine spare parts, procurement personnel manually

identify suppliers through Google searches and send RFQs through email for quotation requests and follow-ups.

Quotation details and supplier responses are then manually organized in Excel sheets or Google Sheets for comparison and coordination with the sales team. Since historical RFQ records and supplier information are not centrally maintained, retrieving previous sourcing information becomes difficult and time-consuming.

### 1.3 Why is AI a viable solution for this problem?

The identified problem is largely related to repetitive information handling and procurement data organization. Procurement personnel spend considerable time manually extracting information from RFQs, identifying products from images, organizing quotation details, and maintaining procurement records.

AI can support this workflow by assisting in RFQ interpretation, image-based product identification, information extraction, and procurement database organization. An AI-assisted system can help structure procurement information into a searchable format, reduce repetitive manual work, improve RFQ tracking, and support faster retrieval of supplier and quotation history during sourcing activities.

### 1.4 When did you identify this problem, and who confirmed it was real?

The problem was identified during the initial phase of the internship while handling live RFQs, supplier sourcing activities, quotation comparisons, and follow-up coordination tasks. The repetitive nature of supplier searching, manual RFQ interpretation, and fragmented procurement information handling became consistently visible during daily operations.
The issue was further discussed with Mr. Sayeem Salamani, Supply Chain Manager, who confirmed that repetitive sourcing activities, lack of procurement information organization, and supplier tracking difficulties are ongoing operational challenges within the current procurement workflow.

## Section 2 — Your AI Solution

### 2.1 What did you build? Describe the solution in plain language.

The proposed solution developed during the internship is titled **ProcureIQ**, an AI-assisted procurement information management system designed to support RFQ interpretation and procurement data organization at Negbuy.

The system is intended to address the operational inefficiencies caused by manual RFQ interpretation and the absence of a centralized procurement information repository. At present, procurement personnel receive RFQs in multiple formats including screenshots, PDFs, Excel sheets, product photographs, technical specification documents, email attachments, and text messages. Each RFQ must be manually interpreted to identify the manufacturer, part number, technical specifications, and quantity before supplier sourcing activities can begin.
ProcureIQ is designed to simplify this process by reading RFQs directly in whatever format they are uploaded. Once the RFQ is uploaded, the system extracts key procurement information including:

- Manufacturer name
- Part number
- Product description
- Technical specifications
- Quantity required
- Product category

After extraction, the information is converted into a structured format and stored inside a centralized procurement database. The database is designed to support future retrieval and procurement reference activities.
The system also includes a search functionality through which procurement personnel can retrieve historical RFQ records using:

- Manufacturer name
- Part number
- Product keyword
- Technical reference

For example, if a procurement executive searches for "Siemens," the system should display all previously stored Siemens-related RFQs and associated procurement information. Similarly, if a specific part number is searched, the system should retrieve all procurement records related to that part number.

The primary objective of ProcureIQ is not procurement automation, but procurement information management and retrieval support. The system is intended to reduce repetitive sourcing effort, improve procurement memory retention, and support faster procurement coordination activities.

### 2.2 What tools or platforms did you use to build it?

| Tool / Platform Used | What It Was Used For |
|---|---|
| Bolt.new | Building and deploying the web application prototype |
| React.js | Front-end application development |
| JavaScript | Application logic and data processing |
| HTML/CSS | User interface design and styling |
| Claude AI | Prototype development support and debugging |
| ChatGPT | Solution design, workflow logic, testing support, and documentation |
| Browser Storage | Storing RFQ records within the prototype database |
| Rule-Based Extraction Engine | Extracting procurement information from RFQs without paid APIs |
| CSV Export Module | Exporting procurement records for analysis and reporting |

### 2.3 What data or documents did the solution use?

The solution used RFQs received during procurement sourcing activities at Negbuy. These included screenshots, product photographs, PDF files, Excel sheets, technical specification documents, email attachments, and plain text RFQ messages. Sample procurement RFQs related to industrial spare parts, automation components, turbine equipment, and supplier quotations were analyzed during the development stage. The solution also utilized manually interpreted RFQ data for testing extraction accuracy and validating the procurement information structure.

### 2.4 What does the solution produce as its output?

The solution produces structured procurement information from unstructured RFQ inputs. The output includes manufacturer name, part number, quantity, technical specifications, product description, and RFQ category. This information is automatically arranged into a centralized database format using Google Sheets. The system also supports basic search and retrieval functionality, allowing users to search previous RFQs using manufacturer names, part numbers, or product descriptions. The output helps reduce repetitive manual interpretation and improves procurement data accessibility.

### 2.5 What are the limitations of your solution?

The current version of ProcureIQ is limited to procurement information extraction and database organization. The system does not currently automate supplier sourcing, quotation comparison, supplier communication, or procurement decision-making activities.

The project also depends on the quality and readability of incoming RFQ formats. Poor image quality, incomplete technical specifications, or unclear product references may reduce extraction accuracy.

At the current internship stage, the system remains a prototype-level solution and has not yet been deployed at organizational scale. Historical procurement data access is also limited because the internship primarily relies on live RFQs received during the internship period.

The current prototype is additionally dependent on external AI APIs and internet-based extraction systems. Full organizational deployment would require:

- Secure database integration
- Internal procurement data permissions
- Larger RFQ training datasets
- Better retrieval architecture
- Dedicated system hosting infrastructure

Future development may include quotation tracking, supplier history mapping, supplier scoring systems, and procurement analytics dashboards, but these capabilities remain outside the current project scope.

# ProcureIQ

## 1. Solution Logic

The system accepts RFQs in multiple formats including screenshots, product photographs, PDFs, Excel files, and plain text. ProcureIQ extracts key information such as manufacturer, part number, quantity, product category, and technical specifications. The extracted information is then organized into a centralized procurement database where users can search, filter, retrieve, and export records.

## 2. Prototype / Working Model

A functional prototype was developed using React-based web application architecture. The prototype consists of:

- RFQ Upload Module
- AI Extraction Module
- Procurement Database Module
- Search and Filter Functions
- Export Functionality
- Dark Mode / Light Mode Interface

The prototype demonstrates the complete workflow from RFQ upload to database storage and retrieval.

## 3. Build Notes

The prototype was developed using:

| Component | Technology Used |
|---|---|
| User Interface | React.js |
| Application Logic | JavaScript |
| Styling | HTML/CSS |
| Development Platform | Bolt.new (browser-based app builder) |
| Design Support | Claude AI & ChatGPT |
| Data Storage | Browser Storage (Prototype Database) |

The system was designed as a proof-of-concept for future organizational deployment.

**How It Was Built**

| Item | Detail |
|---|---|
| **Platform used** | Bolt.new (browser-based app builder) |
| **Frontend framework** | React.js |
| **AI extraction engine** | Free extraction mode (rule-based + pattern matching, no paid API required) |
| **Database** | In-app structured database with 15 real Negbuy RFQ records pre-loaded |
| **Deployment** | Live hosted at https://react-app-creation-74d5.bolt.host |

## 5. Input–Output Flow

**Input**

- Product photographs
- Screenshots
- RFQ text
- PDF files
- Excel spreadsheets
- Email content

**Processing**

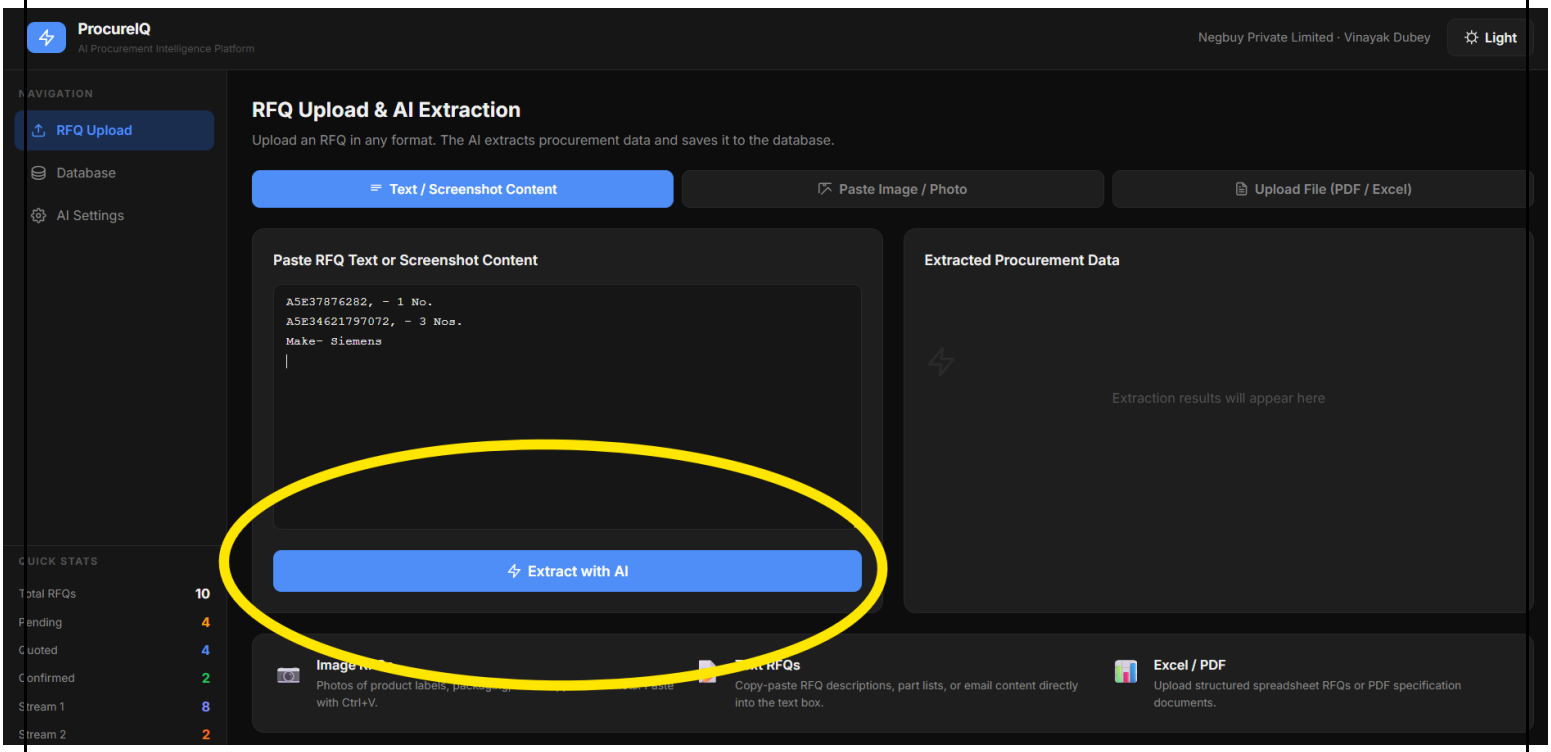
- RFQ interpretation
- Data extraction
- Field classification
- Record generation

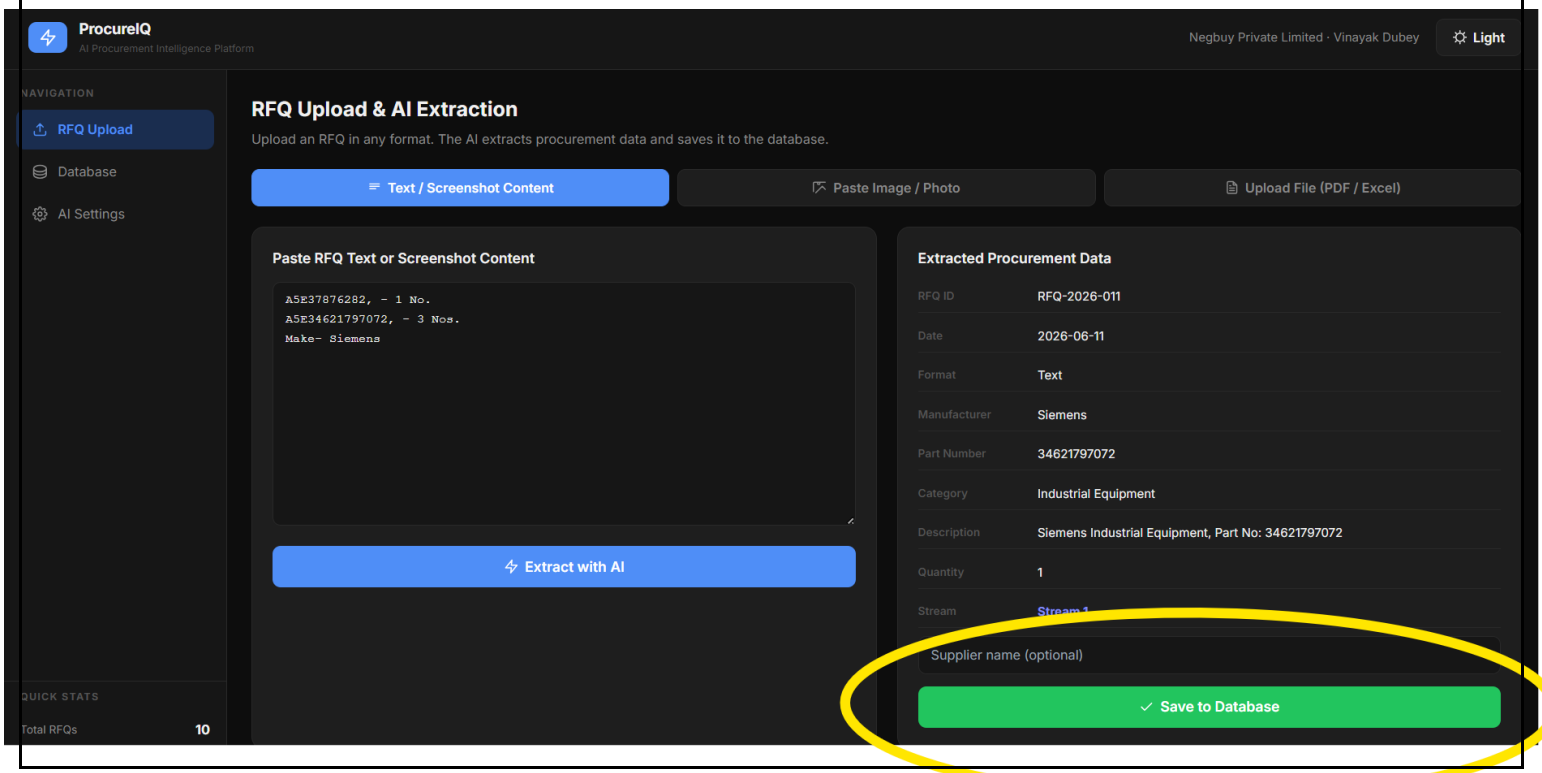
**Output**

- Manufacturer
- Part Number
- Quantity
- Product Category
- RFQ Format
- Status
- Supplier Information
- Searchable Database Record

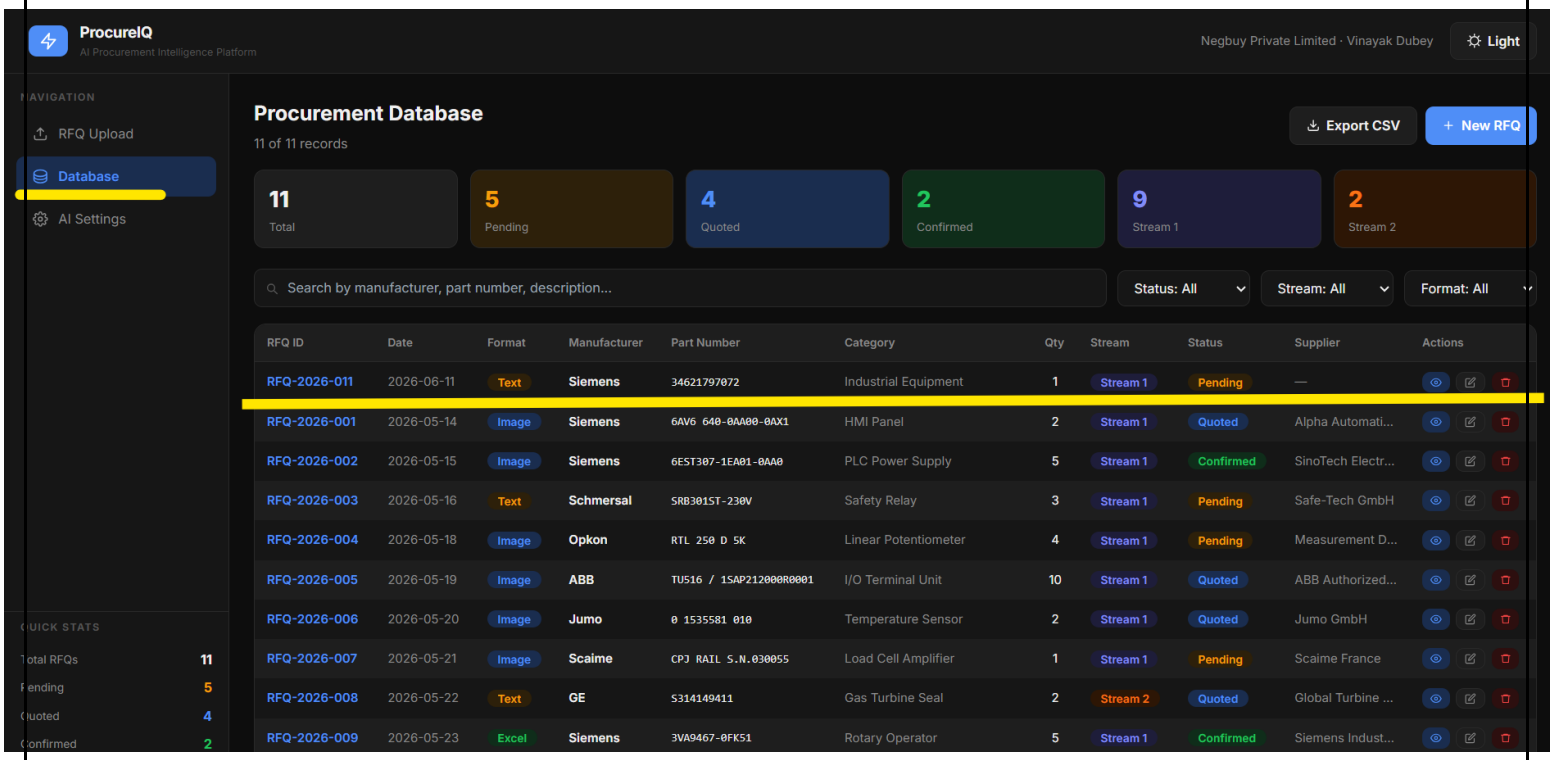
**Example:**

**Input:**
A5E37876282, - 1 No.
A5E34621797072, - 3 Nos.
Make- Siemens



**Output:** When you go to Database option, you can see the 1st entry. The AI analyzed and entered accordingly.




## 6. Basic Testing Evidence

Testing was conducted using sample RFQs collected during the internship. The system was tested using RFQs received in text, screenshot, image, PDF, and spreadsheet formats.

The prototype successfully:

- Accepted multiple RFQ formats
- Generated structured procurement records
- Stored records in a centralized database
- Enabled search and filtering functionality
- Allowed export of procurement data

The testing confirmed the feasibility of using AI-assisted procurement information management to improve RFQ handling and procurement data organization at Negbuy.